



Full length article

ADAP: Adaptive & Dynamic Arc Padding for predicting seam profiles in Multi-Layer-Multi-Pass robotic welding

He Wang^a, Sen Li^a, Xiaobo Liu^b, Chengxiao Dong^a, Fang Wan^a,*

^a School of Design, Southern University of Science and Technology, 1088 Xueyuan Road, Shenzhen, 518055, Guangdong, China

^b Department of Mechanical and Energy Engineering, Southern University of Science and Technology, 1088 Xueyuan Road, Shenzhen, 518055, Guangdong, China

ARTICLE INFO

Dataset link: <https://github.com/ancorasir/WeldingADAP>

Keywords:

Multi-Layer-Multi-Pass welding
Welding simulation
AI for science

ABSTRACT

Welding thick metal plates using Multi-Layer-Multi-Pass (MLMP) techniques demands precise control over the weld seam profile as it evolves during the cooling process. In MLMP welding, typically executed with Gas Metal Arc Welding (GMAW) and shielding gas protection, the continuous deposition of weld beads results in dynamic changes to the seam geometry. These challenging traditional robotic welding systems rely on static models. To ensure high-quality joints, real-time adaptation of welding paths requires accurate predictions of weld bead geometry, which in turn guide the estimation of welding positions for adaptive trajectory planning. In this study, we introduce the Adaptive & Dynamic Arc Padding (ADAP) framework. This novel data-driven approach integrates deep learning with an innovative arc-based representation of weld bead profiles. By representing the weld bead geometry through image-derived boundaries and primitive arc parameters (arc center and radius), ADAP establishes a direct link between welding parameters and the evolving weld seam profile. Utilizing datasets generated from Flow-3D simulations of the MLMP process, our framework achieves high-accuracy, real-time predictions: welding positions are estimated within 0.025 s (with an average error of approximately 1.5 mm), and weld seam profiles are predicted in 15 ms, with the arc-based geometric parameters accurately estimated (average errors of 0.73 mm in arc center position and 0.66 mm in radius). This practical approach enhances the efficiency and quality of MLMP robotic welding and contributes to advances in data-driven modeling and intelligent control in manufacturing, paving the way for autonomous welding systems.

1. Introduction

Thick-plate welding, defined as the joining of metal plates typically 20 mm or thicker, is a critical process in the aerospace, shipbuilding, and heavy engineering industries. The Multi-Layer-Multi-Pass (MLMP) welding technique involves sequentially depositing multiple weld bead passes to build up the joint, as illustrated in Fig. 1. This approach has emerged as an effective solution for fabricating large-volume, high-strength structures. In MLMP welding, Gas Metal Arc Welding (GMAW) is widely employed because it uses a continuous solid wire electrode and shielding gas (commonly an argon-carbon dioxide blend) that stabilizes the arc and prevents oxidation. This method facilitates the controlled filling of extensive joint volumes while minimizing distortion and residual stress, ensuring full penetration and robust weld integrity. For instance, Q355 structural steel is frequently selected for its strength and weldability in large-scale applications.

A significant challenge in MLMP welding lies in the dynamic evolution of the weld seam profile during cooling. As each bead solidifies, its geometry changes, which in turn affects subsequent deposition.

Real-time monitoring and adjustment of the welding torch position are crucial for compensating for these variations and maintaining consistent welding quality. Recent advances in the online detection of weld bead profiles (He et al., 2020) and autonomous welding position decision-making (He et al., 2024b, 2021) have underscored the need for predictive models that can adaptively control the welding process. For example, metallographic observations of actual weld cross-sections in Fig. 1a show that common bead profiles usually exhibit an arc-shape, as illustrated in the two red dashed lines on top. However, as they were deposited on the previous layer, the fusion boundaries are clearly observable, which melted the welding bead profiles of the previous layer (the two red dashed lines in the middle), as marked in dashed blue lines.

Traditional approaches to predicting weld bead geometries have largely relied on empirical models using simplified geometric descriptors such as width, height, and area (Suryakumar et al., 2011; Li et al., 2018). Common shape approximations, such as parallelograms or trapezoids (Madsen et al., 2002; Zhang et al., 2011; Yang et al.,

* Corresponding author.

E-mail address: wanf@ieee.org (F. Wan).

<https://doi.org/10.1016/j.aiopen.2025.10.003>

Received 25 October 2024; Received in revised form 15 October 2025; Accepted 16 October 2025

Available online 21 October 2025

2666-6510/© 2025 The Authors. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

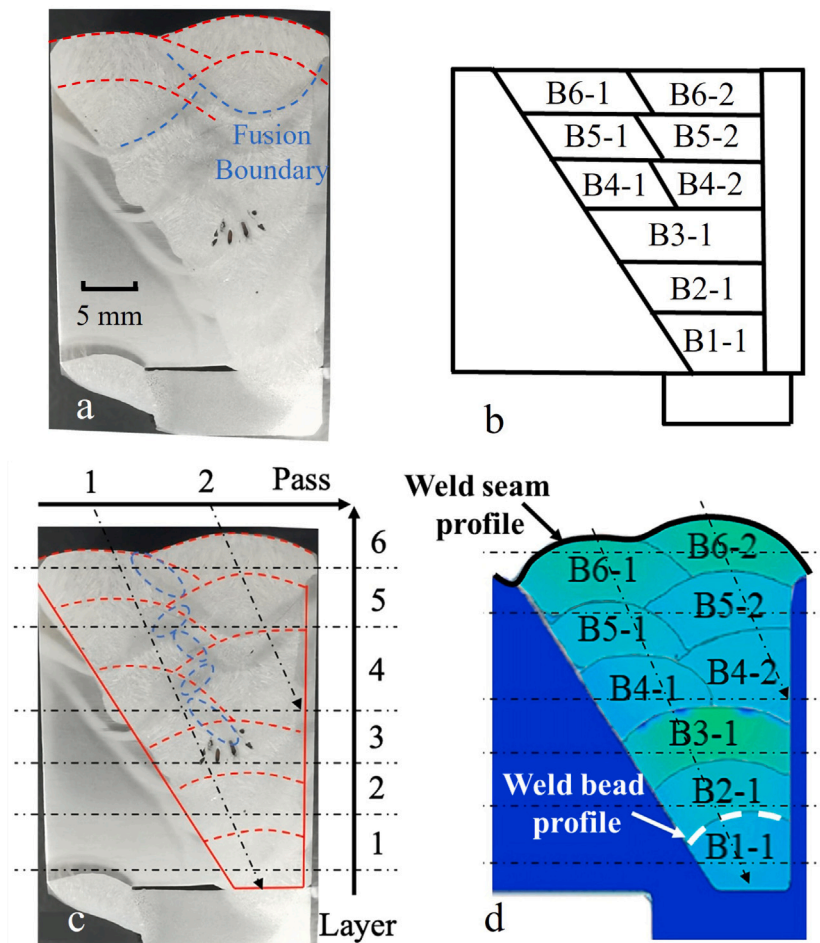


Fig. 1. Modeling weld seam profiles of the Multi-Layer-Multi-Pass (MLMP) welding technique for joining thick plates. (a) Cross-sectional view of two thick plates welded with MLMP techniques; (b) Markings for arc-based weld seam profile indicating MLMP process; (c) Classical model of weld seam profile using parallelograms; (d) Flow-3D simulation of the same welding procedure aligns with our proposed arc-based weld seam profile.

2014), as illustrated in Fig. 1b, often fail to capture the smooth, convex, and continuously evolving cross-sectional profiles of weld beads, especially under dynamic MLMP welding conditions. Our observation shows that the bead profiles are better approximated using circular arcs, as shown in Fig. 1c. Although the precise boundaries of each bead are partially obscured due to fusion and penetration, they can be inferred by analyzing the fusion boundary and deposition sequence, allowing for an arc-based modeling approach to provide a full description. This geometric parameterization offers a more accurate and compact representation compared to traditional models, aligning closely with the Flow-3D simulation results shown in Fig. 1d, thereby establishing a stronger link between simulated and real-world weld bead profiles.

Recent developments in Artificial Intelligence for Science (AI4Science) have paved the way for a data-driven approach to modeling welding processes. Deep learning techniques, supported by high-quality training datasets, have shown promise in enhancing predictive accuracy while reducing reliance on oversimplified empirical assumptions (Akbari et al., 2022; Matsushita et al., 2021; Raouf et al., 2022; He et al., 2024a). Although physical experiments remain indispensable, their high cost and time consumption have led researchers to leverage numerical simulations as an efficient alternative (Ma et al., 2024; Hemmasian et al., 2023). Simulation tools not only capture intricate thermal and mechanical phenomena during welding (Lu et al., 2021) but also offer comprehensive insights into fluid behavior beneath the weld seam, closely aligning with large-scale geometric patterns observed in practice (Wang et al., 2020; Lee et al., 2018; Shin and Boo, 2022).

Motivated by these advances, this paper proposes a novel approach of Adaptive & Dynamic Arc Padding (ADAP), which integrates deep learning with an arc-based geometric model to predict real-time welding positions and weld seam profiles. As illustrated in Fig. 2, ADAP leverages simulation data generated using Flow-3D to train two complementary neural networks: *BeadProfileNet*, which predicts weld bead geometry based on the current weld seam profile and the position of the molten droplet, and *WeldPoseNet*, which estimates the optimal welding torch position. This design-centric representation, utilizing primitive arc geometries, enhances prediction accuracy and enables adaptive adjustments that compensate for the cooling dynamics of the weldment. The contributions of this work are listed as follows:

- Simulated MLMP robotic welding processes and validated the simulation results against experimental data, constructing a comprehensive dataset for MLMP welding.
- Introduced an arc-based modeling approach for accurately representing weld bead profiles in MLMP welding.
- Developed *BeadProfileNet*, a deep learning model that dynamically predicts weld bead geometric parameters from weld seam profiles and molten droplet positions.
- Developed *WeldPoseNet*, a deep learning model that estimates welding positions from weld seam images, facilitating real-time adaptive trajectory planning.

The remainder of the paper is organized as follows. Section 2 details the simulation environment and the architecture of the proposed

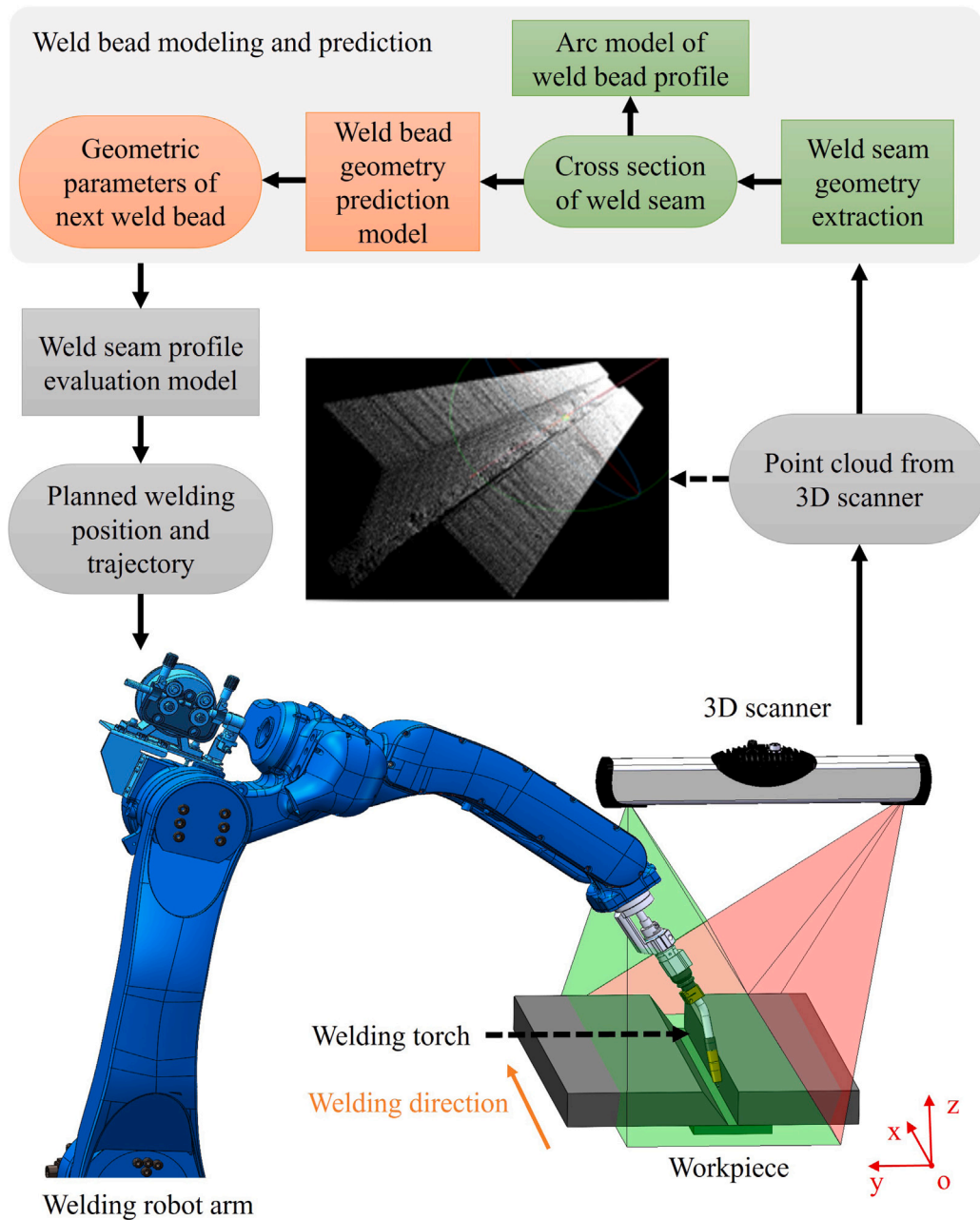


Fig. 2. The workflow of applying Adaptive & Dynamic Arc Padding (ADAP) in Multi-Layer-Multi-Pass robotic welding.

learning-based models. Section 3 presents experimental and simulation results that validate our approach. Finally, we discuss the limitations of the current work and outline potential directions for future research.

2. Materials and methods

To visually summarize the overall architecture and data flow of the proposed ADAP framework, a comprehensive workflow diagram is provided in Fig. 3. This figure illustrates the interplay between the three core components: dataset generation, model development, and framework application, setting the stage for the detailed methodological explanations that follow.

2.1. Robotic welding via the MLMP technique

We conducted an MLMP robotic welding experiment to benchmark the welding simulations in this study. The workpiece consists of two

base plates made of Q355 structural steel, each with a thickness of 23 mm and a backing plate of 6 mm thick. The groove angle between the two base plates was 35°. The shape and relative positions of the workpieces are illustrated in Fig. 4, where the coordinate origin is defined as the midpoint of the lowest plane of the weld seam. The shielding gas employed was Ar–20%CO₂ with a flow rate of 20 L/min, ejected from a nozzle with a 1 cm radius. The filler wire used was ER50-6, with a diameter of 1.2 mm.

The torch was maintained at approximately 80° during each bead welding process, moving linearly along the welding direction at a specified speed. The parameters were carefully determined in the MLMP welding process based on pre-planned bead arrangement, with current, voltage, and wire feed speed matched according to established relationships. A higher current (approximately 300 A) was selected for beads requiring a larger deposition volume. Additionally, the welding positions were strategically chosen at lower points in the groove to create flat surfaces beneficial for subsequent passes. Then, experienced

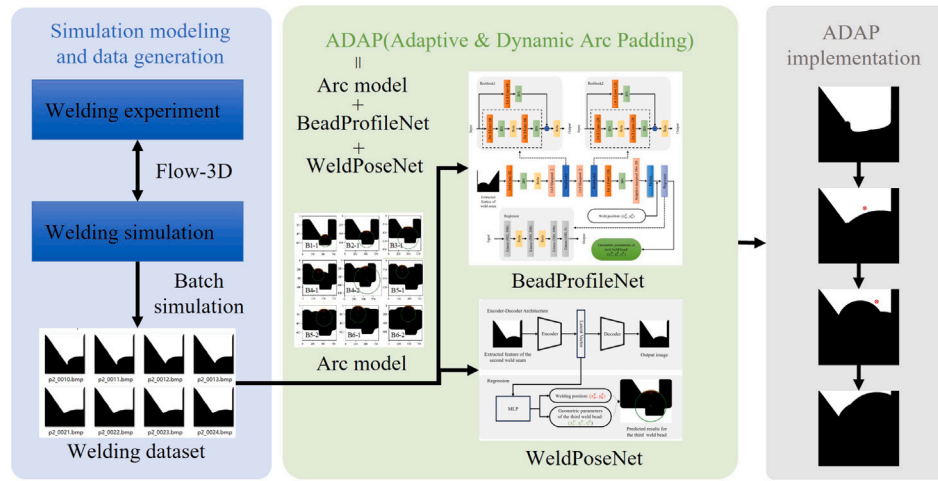


Fig. 3. Overview of the ADAP framework and workflow.

Table 1

Welding parameters for the six-layer nine-pass single bevel groove welding.

Weld bead	Start position (mm)	End position (mm)	Welding speed (mm/s)	Welding current (A)	Welding voltage (V)
B1-1	(0, 0, 3)	(50, 0, 3.0)	5	270	28
B2-1	(0, 1.2, 7.8)	(50, 1.2, 7.8)	4.5	300	31
B3-1	(0, 0.35, 11.8)	(50, 0.35, 11.8)	4	300	31
B4-1	(0, 9.2, 13.8)	(50, 9.2, 13.8)	4.5	300	31
B4-2	(0, 1.0, 16.8)	(50, 1.0, 16.8)	3	300	31
B5-1	(0, 11.7, 18.3)	(50, 11.7, 18.3)	5	300	31
B5-2	(0, 2.2, 21.8)	(50, 2.2, 21.8)	4.5	300	31
B6-1	(0, 12.0, 23)	(50, 12.0, 23.0)	3	300	31
B6-2	(0, 2.8, 25)	(50, 2.8, 25.0)	5	300	31

2.2. Simulating the MLMP robotic welding

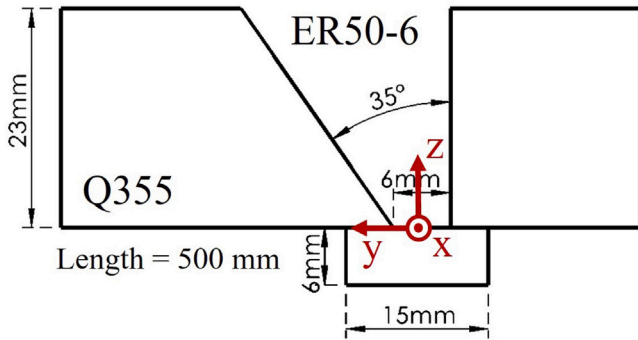


Fig. 4. The single bevel groove's cross-section used in the MLMP robotic welding task.

welding professionals from our industrial collaborator (China Construction Science and Industry Steel Co., LTD.) fine-tuned these welding parameters based on their expertise and practical considerations to ensure the welding quality, aiming to prevent or minimize the occurrence of welding defects. The criteria for high-quality weld beads include the uniform height of the weld bead, full penetration on the back side, and good fusion between the weld bead and sidewalls. Based on the size of the weld seam, the weld seam was divided into nine beads in Fig. 1. The experimental welding parameters for each bead are set in Table 1, where Bi-j represents the j th weld bead in the i th layer in the multi-layer multi-pass (MLMP) welding process.

This study simulates the molten pool using the commercial software Flow-3D.¹ It implements the fundamental governing equations: mass conservation, momentum conservation, and energy conservation equation (Xu et al., 2018). The weld pool surface is considered a free surface, and the Volume of Fluid (VOF) technique is employed to track its movement (Hirt and Nichols, 1981; Dai and Gu, 2014). The Mass Conservation equation is

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) = S_m. \quad (1)$$

The Momentum Conservation equation is

$$\rho \left(\frac{\partial \vec{V}}{\partial t} + \vec{V} \cdot \nabla \vec{V} \right) = -\nabla p + \mu \nabla^2 \vec{V} - \frac{\mu \vec{V}}{K} + \rho \vec{g} \beta (T - T_{ref}) + \vec{F}. \quad (2)$$

The Energy Conservation equation is

$$\frac{\partial(\rho H)}{\partial t} + \vec{V} \cdot \nabla(\rho H) = \nabla \cdot (k \nabla T) + S_v, \quad (3)$$

where ρ is density, $\vec{V}$ is the velocity vector, p is pressure, μ is viscosity, t is time, $\vec{g}$ is the gravitational vector, T_{ref} is the reference temperature, β is the thermal expansion coefficient, H is enthalpy, K is the resistance coefficient in porous media, $\vec{F}$ is the electromagnetic force vector, and S_m and S_v are the mass and heat source terms, respectively. The metal and gas fluids are treated as incompressible Newtonian laminar flows to simplify calculations (Xu et al., 2016).

¹ <https://www.flow3d.com/>

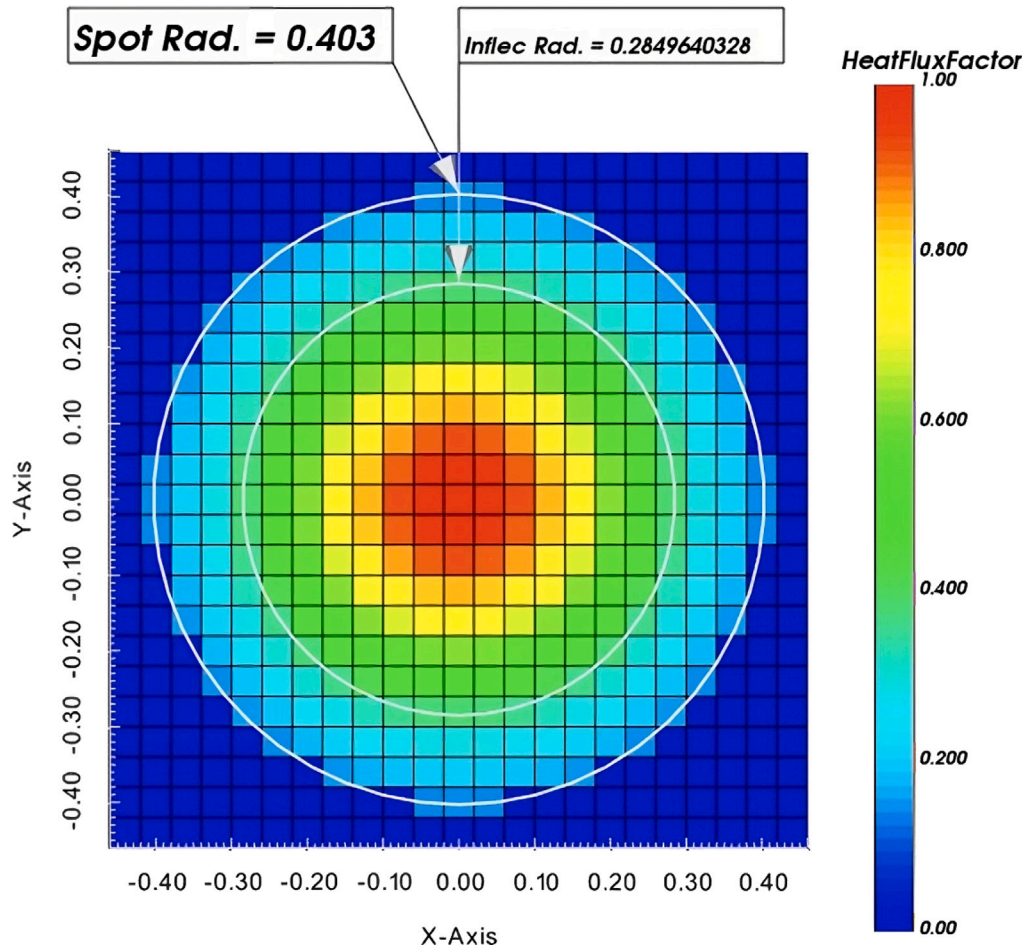


Fig. 5. The Gaussian heat source used in the simulation and its heat flux distribution.

2.2.1. The heat source model

The arc heat input is modeled as a Gaussian surface heat source applied to the surface of the melt pool. Its rationality has been verified in [Cho and Na \(2009\)](#) with the following expression:

$$q = \frac{\mu UI}{r_0^2 \pi} \exp \left\{ - \left(\frac{\sqrt{2}r}{r_0} \right)^2 \right\}, \quad (4)$$

where μ is the arc efficiency, U is the welding voltage (V), I is the welding current (A), r_0 is the radius of the heat source (mm), r is the radial coordinate (mm), q is the heat flux density (W/mm²). The surface heat flux at radius r is q , and materials outside r_0 will not absorb any heat.

Based on previous work ([Cho et al., 2013](#)), the arc efficiency was set to 0.56. The heat source radius parameter of the Gaussian heat distribution varies with the change of pulse current, and the heat source radius can be calculated according to its empirical formula ([Lu and Kou, 1988](#))

$$r_0 = \sqrt{2} \times 0.533 I^{0.2941}. \quad (5)$$

The heat flux distribution is shown in [Fig. 5](#).

2.2.2. The molten droplet model

During the calculation process, the molten droplet is assumed to behave as a high-temperature liquid metal flowing into the molten pool from a designated area above it ([Xu et al., 2018](#)). In this article, we simulate the welding results of different welding positions by varying the initial positions of the droplets.

The temperature of the droplet influences the welding temperature field and the geometry of the final weld bead profile. According to [Cho and Na \(2009\)](#), the molten droplet's initial temperature was 2200 K. When the diameter of the wire is 1.2 mm, the welding current is 300 A. The wire feeding speed is 11.6 m/min, and the radius of the molten droplet is calculated to be 0.6 mm, according to the work of [Lin et al. \(2001\)](#). Its velocity consists of a vertical component only, with a falling velocity of approximately 1.5 m/s.

2.2.3. The pressure model

In the arc welding process, the molten pool experiences a slight deformation along its top surface due to the arc pressure generated by the substantial flow of arc plasma. Referring to the previous work ([Xu et al., 2018](#); [Cao et al., 2004](#)), the distribution of arc pressure can be described as

$$P_{arc}(t) = \frac{P_{total}}{\sigma_p^2 \pi} C_{time}(t) \exp \left\{ - \left(\frac{r}{\sqrt{2}\sigma_p} \right)^2 \right\}, \quad (6)$$

where P_{arc} is the arc pressure density function, P_{total} is the total arc force, C_{time} is the square wave period function with a function value of 0 or 1, the period is the generation interval of the molten droplet, the function value is 1 when the molten droplet is first generated and is zero after the generation interval of a molten droplet, r is the radial coordinate and σ_p is the Gaussian distribution parameter. According to previous studies, the empirical equation for the total arc force is expressed as ([Cho and Farson, 2007](#))

$$P_{total} = -0.04017 + 0.0002553 I. \quad (7)$$

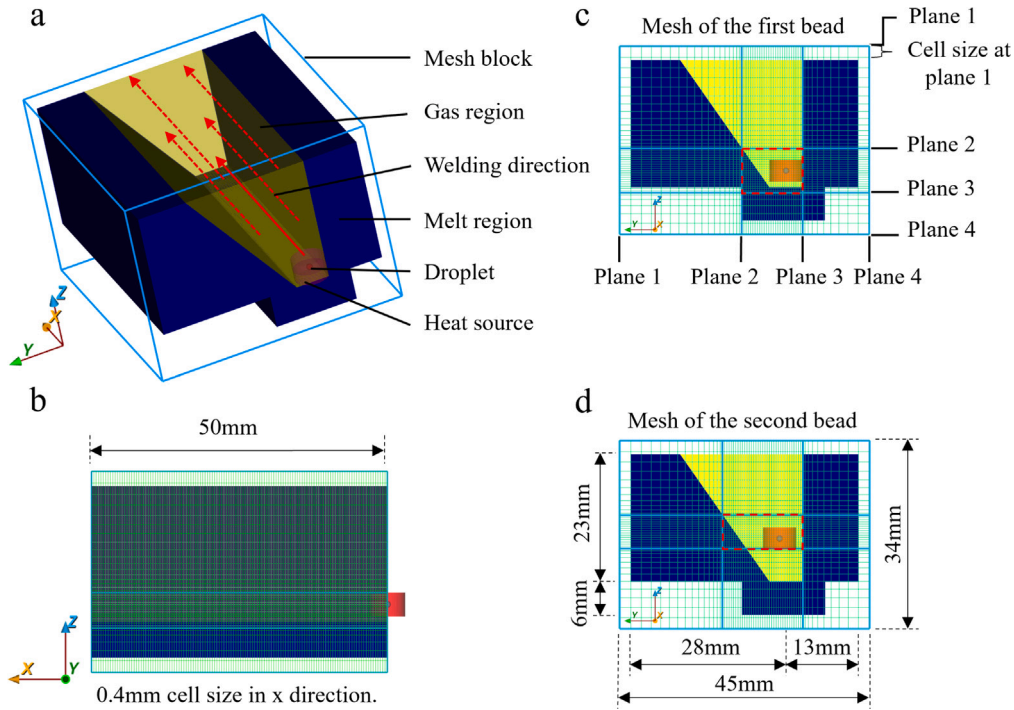


Fig. 6. Simulation setup in Flow-3D. (a) Computational domain. (b) Cell size in the x direction. (c) Mesh plane set for the first bead. (d) Mesh plane set for the second bead.

The Gaussian distribution parameters of the arc force are

$$\sigma_p = 1.4875 + 0.00123 \cdot I(\text{mm}). \quad (8)$$

2.2.4. Meshing and material parameters

The workpiece's dimensions are 41 mm in width and 29 mm in height. The entire computational volume is sized at 50 mm $\times$ 45 mm $\times$ 34 mm, which is slightly larger than the workpiece, to simulate the heat dissipation of the workpiece through the adjacent void region. The volume's length is 50 mm, shorter than the actual workpiece, but it is long enough to form a uniform weld bead along the x -axis and yet not computationally demanding. The height of the heat source beam is 4 mm, moving along the welding direction in the positive x -axis direction. Molten droplets are generated at regular intervals in the center of the heat source beam. Subsequently, these droplets descend at a certain velocity along the negative direction of the z -axis. The relevant simulation setup is shown in Fig. 6a. Along the x -direction of the computation domain, the cell size is set to 0.4 mm, as shown in Fig. 6b.

Variable mesh sizes were implemented in the simulation to optimize simulation efficiency. The cell size in mesh planes 1 and 4, which demarcate the boundaries of the computational domain, was set to 2 mm, a size conducive to reducing computational load. In contrast, mesh planes 2 and 3, encompassing the periphery of the weld bead, were assigned a smaller cell size of 0.4 mm. This finer mesh resolution in planes 2 and 3 was crucial for accurately capturing the intricate dynamics of the weld bead formation. The judicious use of coarser meshes in areas less critical to weld bead formation facilitated a reduction in computational time while maintaining an acceptable level of simulation accuracy. The mesh division for the first and second weld beads simulation is illustrated in Figs. 6c&d. The values of the material thermo-physical parameters used to simulate the molten pool behavior are listed in Table 2.

Table 2

Thermo-physical properties of workpiece and filler wire used in the simulation.

Parameter	Unit	Numeric value	
		Q355	ER50-6
Density, 298 K	g/cm ³	7.81	7.8
Density, 2073 K	g/cm ³	6.7	6.75
Special heat	J/g/K	0.48	0.45
Thermal conductivity	W/m/K	48	40.62
Viscosity	mPa s	4.82	4.29
Surface tension	N/m	1.2	1.2
Solidus temperature	K	1633	1100
Liquidus temperature	K	1773	1200
Latent heat of fusion	J/kg	2.61×10^5	2.77×10^5

2.2.5. Boundary conditions

In Fig. 6a, the upper surface of the mesh model is set as the pressure outlet boundary. The boundary conditions should satisfy the following equations

$$P_{\text{outlet}} = P_0, \quad (9)$$

$$T_{\text{outlet}} = T_0, \quad (10)$$

where P_0 and T_0 represent the ambient temperature and pressure, respectively. For the energy boundary conditions of the free surface, arc heat flux, and heat loss from convection, radiation, and evaporation are considered. The energy balance model for a free surface is (Cho and Na, 2015)

$$k_t \frac{\partial T}{\partial n} = q - q_{\text{conv}} - q_{\text{rad}} - q_{\text{evap}}, \quad (11)$$

where k_t is the thermal conductivity, T is the surface temperature, ∂n is the normal vector of the free surface, q is the arc heat, q_{conv} is the convection heat dissipation, q_{rad} is the radiation heat dissipation, and q_{evap} is the evaporative heat dissipation. The shielding gas from the torch head section during welding protects the welding area from oxygen and water vapor. In Fig. 6, the yellow area represents $Ar - 20\%CO_2$ shielding gas with a flow velocity of approximately 106 cm/s, in which a dynamic pressure is applied to the molten metal interface through a

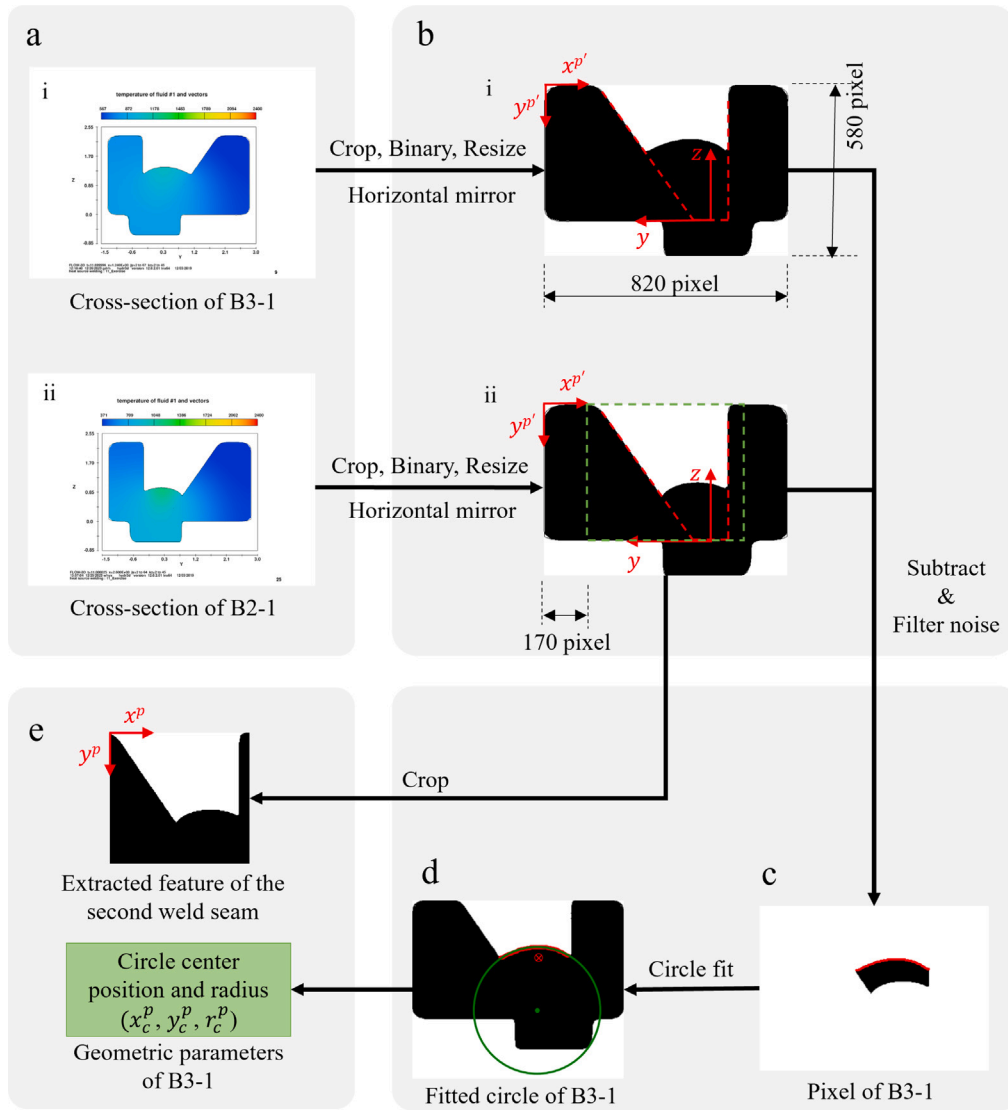


Fig. 7. The workflow of weld bead profile modeling. (a) Cross-section of weld seams in simulation. (b) Binary cross-section images. (c) Extracted weld bead profile. (d) Circle fitting of weld bead section contour. The green dot is the center of the circle, and the red $\otimes$ is the welding position. (e) The final image of the weld seam and geometric parameters of the weld bead.

single-fluid model, and the dynamic pressure formula is (Flow Science Japan, 2023):

$$P_{gas} = \frac{1}{2} \rho_{gas} v_{gas}^2 \cos \theta, \quad (12)$$

where P_{gas} is the dynamic pressure of the shielding gas, ρ is the density of the shielding gas, v_{gas} is the flow velocity of the shielding gas, and θ is the angle between the velocity of the shielding gas and the normal vector of the free surface. The pressure boundaries on the free surface are (Cho and Na, 2015):

$$p = P_{arc} + P_{gas} + \frac{\gamma}{R_c}, \quad (13)$$

where p is the free surface pressure, γ is the surface tensor, and R_c is the radius of surface curvature. All other boundaries, including the rear, front, and sides, are symmetrical.

2.3. Arc-based modeling of welding bead profiles

Based on the work by Xiong et al. (2013), the arc model exhibits greater accuracy than the parabola model and cosine model for a single bead profile when the ratio of the wire feed rate to the welding speed

exceeds 12.5. In this experiment, the ratio of the wire feed rate to the welding speed is about $11.6 \times 1000 \div 60 \div 6 \approx 32.2$, which is greater than 12.5. Therefore, we adopt the arc model to fit the weld bead profile. It introduces an innovative modeling approach for weld beads on uneven surface deposition and its application in MLMP welding.

The arc model is obtained in three steps: extraction of the weld seam profile, extraction of the weld bead profile, and fitting of the weld bead profile to a circle. This method utilizes a parametric representation of the weld bead section's geometry. As an example, the modeling for the third weld bead profile is illustrated in Fig. 7. Firstly, the y - z cross-sectional images of the weld seam, before and after the completion of the third weld bead, are exported using Flow-3D software, as shown in Figs. 7a(i)&(ii), respectively. These images have dimensions of 1319×791 pixels. A Python program extracts the region of the workpiece by cropping, binarizing, and horizontally mirroring the images. Following this, the dimensions of the extracted image in pixels are resized such that the width is 820 pixels and the height is 580 pixels while maintaining the aspect ratio proportional to the actual workpiece dimensions, which are 41 mm in width and 29 mm in height, as depicted in Fig. 7b. The white pixels in the extracted images represent void regions.

Further, the extracted cross-section of the previous weld seam, shown in Fig. 7b(ii), is cropped to remove the base plate regions. The purpose is to focus on the weld seam region and provide essential boundary information for predicting the next weld bead's profile, as shown in Fig. 7e. This cropping helps reduce the image size and facilitates training for deep learning, as explained in the following subsection.

Lastly, the region of the third weld bead is then obtained by subtracting the extracted weld seam profile before in Fig. 7b(ii) and after in Fig. 7b(i) the completion of the third weld bead. The color of the resultant image is inverted, and black regions with an area of less than 20 are removed as noise. As illustrated in Fig. 7c, the red curve represents the weld bead profile to be fitted using the arc model. The fitting function is defined as

$$height_{pix} - y^{p'} = \sqrt{(r_c^{p'})^2 - (x^{p'} - x_c^{p'})^2} + (height_{pix} - y_c^{p'}), \quad (14)$$

where $height_{pix}$ is the image height in pixels, $x^{p'}$ and $y^{p'}$ are the positions in the pixel coordinate system, as illustrated in Fig. 7b. $x_c^{p'}$, $y_c^{p'}$, and $r_c^{p'}$ are the fitting parameters, representing the center position and radius of the circle.

The fitting error (FE) is given by

$$FE = \frac{1}{n} \sqrt{\sum_{i=1}^n (y_i^{p'} - y_{ai}^{p'})^2}, \quad (15)$$

where $y_i^{p'}$ and $y_{ai}^{p'}$ represent the coordinates of the fitting points and the coordinates of points on the red arc in Fig. 7c, respectively.

The coefficient of determination (R^2) is defined as

$$R^2 = 1 - \frac{\sum_{i=1}^n ((y_i^{p'} - y_{ai}^{p'})^2)}{\sum_{i=1}^n ((\bar{y}_a^{p'} - y_{ai}^{p'})^2)}, \quad (16)$$

where $\bar{y}_a^{p'}$ is the mean of $y_{ai}^{p'}$. By utilizing the `curve_fit` function from the OpenCV (cv2) library in Python, and employing the fitting function in Eq. (14) along with the pixel coordinates of points on the red curve (Fig. 7c) as inputs, we obtain three fitted parameters: $x_c^{p'}$, $y_c^{p'}$ and $r_c^{p'}$. Consequently, we determine the position of the center and the radius of the fitted circle for the third weld bead profile, as illustrated in Fig. 7d. The red $\otimes$ marks the welding positions in the y-z plane, the green circle depicts the fitted circle, and the green dot denotes the center of the fitted circle.

As shown in Fig. 7e, through the workflow of weld bead profile modeling, we obtain critical information that determines the profile of the third weld bead B3-1, namely the weld seam profile before the third weld bead starts as the boundary condition, the position of the molten droplet (used to simulate the welding position), and the resultant weld bead profile represented by geometric parameters (x_c^p , y_c^p , r_c^p).

The coordinate transformation between the two image coordinate systems and the coordinate system used in the welding experiment is as follows

$$w_{scale} = \frac{width_{pix}}{41}, \quad (17)$$

$$h_{scale} = \frac{height_{pix}}{29}, \quad (18)$$

$$x^{p'} = width_{pix} - (y + 13) \times w_{scale}, \quad (19)$$

$$y^{p'} = height_{pix} - (z + 6) \times h_{scale}, \quad (20)$$

$$x^p = x^{p'} - 170, \quad (21)$$

$$y^p = y^{p'}, \quad (22)$$

$$r_c^p = r_c^{p'}, \quad (23)$$

where $width_{pix}$ and $height_{pix}$ represent the number of pixels in the width and height directions of the workpiece picture shown in Fig. 7b, respectively. The workpiece's cross-section width and height are 41 mm

and 29 mm, respectively. w_{scale} and h_{scale} represent the scaling factors for dimension conversion. y and z represent the coordinates in the workpiece coordinate system, as illustrated in Fig. 4. x^p , y^p , and r_c^p are the coordinates of the final image coordinate system illustrated in Fig. 7.

2.4. BeadProfileNet: Learning welding bead profile prediction

As long as the welding parameters are known and fixed, we could simplify the prediction of the following weld bead profile by predicting the geometric parameters (x_c^p , y_c^p , r_c^p) of the next bead with known current weld seam profile as the boundary condition and the position of the molten droplet. Instead of building an explicit analytical model, we employ a deep learning method, as we obtain sufficient training data from simulation (further explained in Section 3.1.2). This direct input-output mapping problem is well-suited for a convolutional neural network (CNN) based on ResNet, which is particularly effective in handling spatial feature extraction tasks. The neural network takes as input an extracted feature of a weld seam image and a welding position, and outputs the corresponding geometric parameters. The network architecture starts with a 3×3 convolutional layer with 32 channels, followed by batch normalization, a ReLU activation function, and a 3×3 max-pooling layer with a stride of 2. Then, there are two Resblocks, with each Resblock's shortcut connection consisting of a convolutional layer and a batch normalization. Between the two Resblocks is a 3×3 max-pooling layer with a stride of 2. Afterward, a 1×1 convolutional layer with 128 channels, batch normalization, and a 30×30 AdaptiveMaxPool2d layer are applied to extract the two-dimensional features of the weld seam image.

These features are then flattened into a one-dimensional vector and combined with the pixel coordinates of the welding position. The pixel coordinate system is illustrated in Fig. 7e. The combined features are then input into a Regressor block consisting of four fully connected layers and two ReLU activation functions. Finally, the output of the Regressor block provides the geometric parameters for the formed weld bead, including the position and radius of the fitted circle (see Fig. 8).

This problem is a regression task for weld bead geometric parameters, so we use the root mean squared error (RMSE) as the loss function to train our model. As the network output consists of position and radius, we use a combined loss as the training loss function, which is the sum of the errors in the position and radius of the fitted circle. The loss is defined as follows:

$$e_c = \sqrt{\frac{(\hat{x}_c^p - x_c^p)^2 + (\hat{y}_c^p - y_c^p)^2}{2}}, \quad (24)$$

$$e_r = |\hat{r}_c^p - r_c^p|, \quad (25)$$

$$loss = e_c + e_r, \quad (26)$$

where $loss$ represents the prediction error, e_c and e_r denote the position error and radius error components of the prediction error, respectively. $(\hat{x}_c^p, \hat{y}_c^p, \hat{r}_c^p)$ is the predicted result of the network, and (x_c^p, y_c^p, r_c^p) is the label shown in Fig. 7e.

2.5. WeldPoseNet: Learning welding position prediction

Experienced welders determine the welding position by visually inspecting the weld seam surface in practical multi-layer, multi-bead welding. Similarly, this study utilizes a deep learning model to predict welding positions based on images of weld seams. Specifically, an Encoder-Decoder architecture is employed to encode the weld seam image into a structured latent space. This approach preserves the essential geometric features of the weld seam profile while effectively filtering out local noise, enhancing the robustness of the welding position prediction. WeldPoseNet, illustrated in Fig. 9, contains two parts: an *encoder-decoder architecture* for feature extraction and a *multilayer perceptron (MLP)* for estimating the welding position and geometric

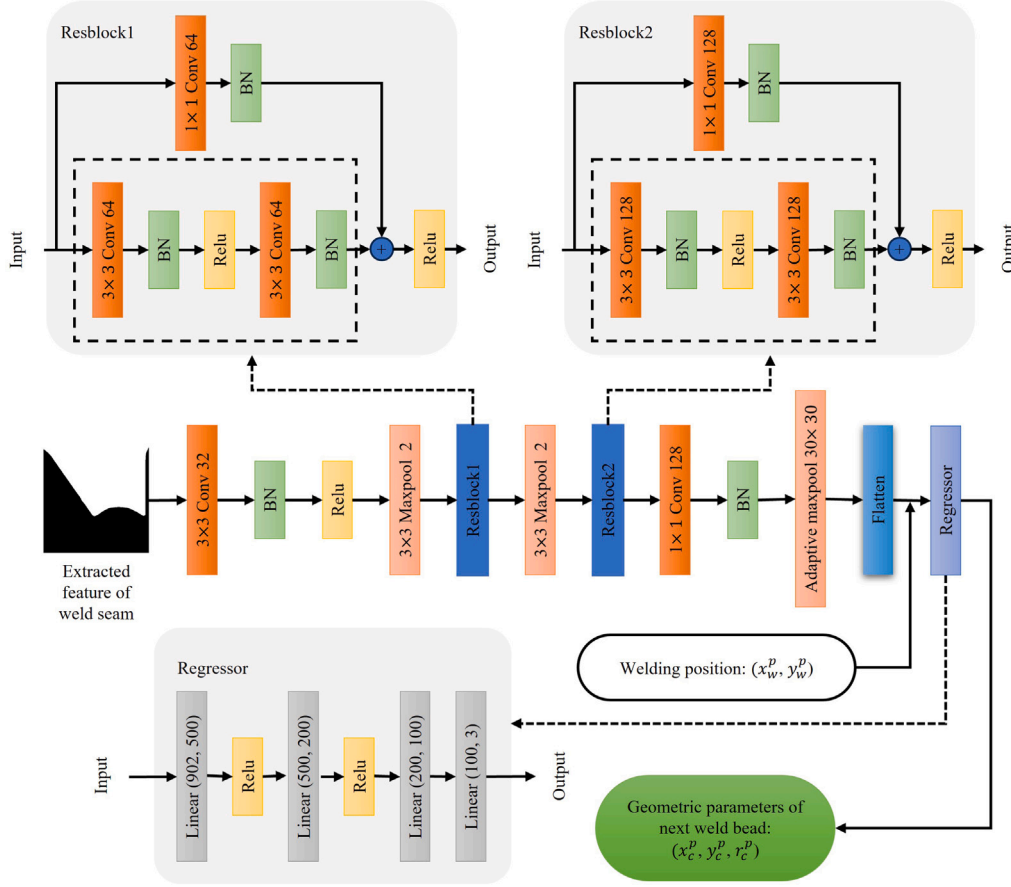


Fig. 8. The network architecture using ResNet blocks. The inputs are the extracted image and welding position, and the output is the parameters of a circle.

parameters in the pixel coordinate system. The input of the WeldPoseNet is an image of the weld seam profile. The encoder–decoder architecture follows the design outlined in Liu et al. (2023) and outputs a 256-dimensional latent vector. The regression model comprises three hidden layers with 200, 200, and 100 neurons, employing the rectified linear unit (ReLU) activation function and batch normalization. The final output is a 5-dimensional vector.

Since WeldPoseNet is primarily designed for welding position prediction, the prediction error of the welding position (e_g) carries greater weight compared to the prediction error of geometric parameters (e_p). Therefore, the loss function for the regression model is defined as follows:

$$e_g = \sqrt{\frac{(\hat{x}_c^p - x_c^p)^2 + (\hat{y}_c^p - y_c^p)^2 + (\hat{r}_c^p - r_c^p)^2}{3}}, \quad (27)$$

$$e_p = \sqrt{\frac{(\hat{x}_p^p - x_p^p)^2 + (\hat{y}_p^p - y_p^p)^2}{2}}, \quad (28)$$

$$\text{loss} = e_p + 0.1 \times e_g, \quad (29)$$

3. Results and discussions

3.1. MLMP welding simulation

3.1.1. Six-layer-nine-pass welding simulation

According to the experimental welding parameters from Table 1, we use Flow-3D software to simulate the welding process. For the first weld bead, the initial temperature of the workpiece is set to 298 K. Due to the smaller length of the workpiece in the simulation compared to its actual dimensions, the heat dissipation of the workpiece in the simulation is

lower than in the real experiment (Deng et al., 2019). Consequently, after each subsequent welding bead, the temperature of the workpiece in the simulation increases more than that in the experiment, leading to some discrepancies. Furthermore, this discrepancy in workpiece temperature affects the geometry of the weld bead, which results in thicker weld beads as the workpiece temperature rises (Manokruang et al., 2021). Typically, the workpiece temperature rises by approximately 100–200 K during welding. To achieve a simulation closer to the welding experiment, the temperature of the fluid region, including the workpiece and the filled portion of the weld seam, is set to 470 K at the beginning of each weld bead simulation after the fourth weld bead.

The three-dimensional results of the six-layer nine-pass simulation are shown in Fig. 10. In the metallographic image of the weld seam from welding experiments shown in Fig. 1a&c, the solid red line represents the initial weld seam boundary, and the profile of the weld bead is positioned above the fusion boundary (indicated by blue dashed lines) of the following weld bead due to the penetration. Each layer consists of two weld beads starting from the fourth layer. After B4-2 is formed, the fusion boundary appears in the middle of the weld seam, as indicated by the bottom blue dashed line. Due to the penetration of B4-2, the profile of B3-1 is slightly higher than the bottom of this fusion boundary. Based on the wire feeding and welding speeds, the proportion of the cross-sectional area of B1-1, B2-2, and B3-1 can be estimated. Subsequently, the approximate positions of the profiles of B1-1 and B2-1 can be determined. Besides, the profile of B4-1 is slightly higher than the top of this fusion boundary. Similarly, the profiles of B4-2, B5-1, B5-2, and B6-1 can be estimated based on the subsequent middle portions of the fusion boundaries. The profile of B6-2 can be directly determined from the image. Finally, the profiles of the weld beads in this experiment are approximately represented by the red dashed line in Fig. 1c. Fig. 1d shows the cross-section of the simulated

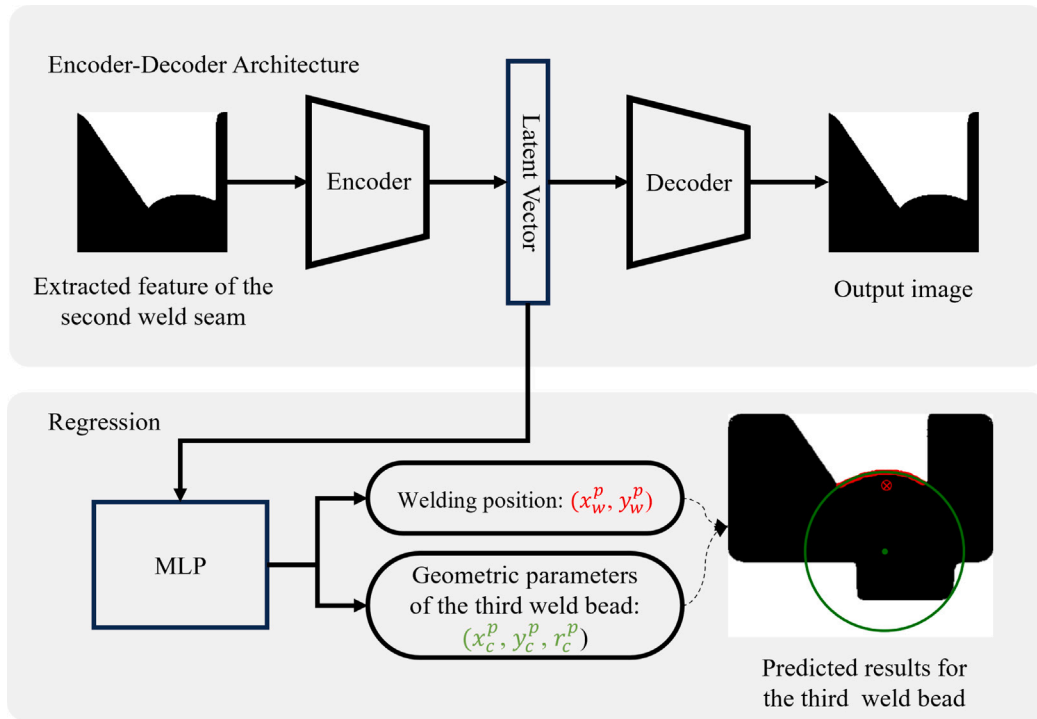


Fig. 9. WeldPoseNet's architecture, involving one resized grayscale image of the weld seam as input, and predicts welding position and geometric parameters of the next weld bead.

weld seam at approximately $x = 18$ mm after completing all nine weld beads. Comparing the weld bead profile images from Zhan et al.'s work on MLMP welding experiments (Zhan et al., 2017) validates our division results of the weld bead profiles. The simulated weld bead profiles generally agree well with the experimental results, as shown in Figs. 1c&d, thus verifying the accuracy of the simulation data.

3.1.2. A simulated MLMP welding dataset

In the MLMP welding process, the first weld bead is typically positioned at the center of the weld seam. The weld beads on the top layer primarily contribute to the aesthetic appearance of the finished workpiece. The quality of MLMP welding mainly depends on the weld beads in the intermediate filling layers. Therefore, we only generated a 7-bead dataset for the intermediate filling weld beads. Specifically, for nine-pass welding of a 23 mm thick plate, we simulated the first backing weld bead and the subsequent six filling weld beads. Since no layer planning was done, these seven weld beads are labeled B1 to B7. In this article, we explored the influence of welding position on weld bead geometry. So, the welding speed, current, and voltage were kept constant in each weld bead, set to 5 mm/s, 300 A, and 31 V, respectively.

The length of each weld bead increased from 5 cm to 10 cm, and the trajectory of the first backing weld bead was a straight line. In contrast, the trajectories of the following six weld beads are all two-segment polylines defined by three points. The position of the droplets varies along the welding direction (x -direction). This allows us to generate more diverse data than with a straight-line trajectory. In this data generation process, the initial weld seam shape and mesh partitioning are the same as the six-layer and nine-pass welding simulation in Section 3.1.1. The welding trajectory details are shown in Table 3, the coordinates of points are in Cartesian coordinates, and the sequence is (x, y, z) . The complete welding trajectory consists of two line segments, calculated by linear interpolation between the start and midpoint, or the midpoint and endpoint.

The length of the workpiece in the x -direction is 10 cm, and the cell size in the simulation is 0.4 mm so that we can extract 250 weld seam

Table 3

Welding trajectory parameters used to generate the simulated dataset.

Weld bead	Start position (mm)	Midpoint position (mm)	End position (mm)
B1	(0, 0, 3.0)	(50, 0, 3.0)	(100, 0, 3.0)
B2	(0, -2.0, 7.0)	(50, 9.0, 8.5)	(100, -2.0, 10.0)
B3	(0, 9.0, 10.0)	(50, -2.0, 12.0)	(100, 9.0, 14.0)
B4	(0, -2.0, 11.0)	(50, 9.0, 14.0)	(100, -2.0, 11.0)
B5	(0, -2.0, 18.0)	(50, 12.0, 18.0)	(100, -2.0, 18.0)
B6	(0, 12.0, 18.0)	(50, -2.0, 18.0)	(100, 12.0, 18.0)
B7	(0, -2.0, 20.0)	(50, 12.0, 21.5)	(100, -2.0, 23.0)

cross-sections in a single weld bead. However, due to the fluidity of the weld pool, the end of each weld bead is not fully formed. Therefore, we excluded the last 50 sections and selected only the first 200 weld seam sections in each weld bead. The total number of the simulated dataset is $1400 = 200 \times 7$. The trajectory of B1 is a straight line, so the cross-sections in B1 are the same; the available number of samples in the dataset is 1200 in B2-B7.

In short, this dataset includes welding positions, profiles of weld beads, and fitted circle parameters. The welding positions are (x_w^p, y_w^p) in the image coordinate system shown in Fig. 7e, and the positions only contain the data in B2-B7. The weld bead profiles contain the data of B1-B6 and are images of 480×450 pixels. The corresponding circle parameters (x_c^p, y_c^p, r_c^p) are calculated from weld bead contours in B2-B7 using the workflow shown in Fig. 7.

3.2. Arc-based modeling of welding bead profiles

The cross-sections of each welding bead are the same along the welding direction in the six-layer, nine-pass welding simulation. So we extract cross-section images at $x = 18$ mm for all weld beads shown in Fig. 7. Following the workflow demonstrated in Fig. 7, we fit circles to the welding bead contours. The fitting results of the cross-sectional contours for each weld bead are shown in Fig. 11. The red line is the contour of the welding bead, the red $\otimes$ is the welding position,

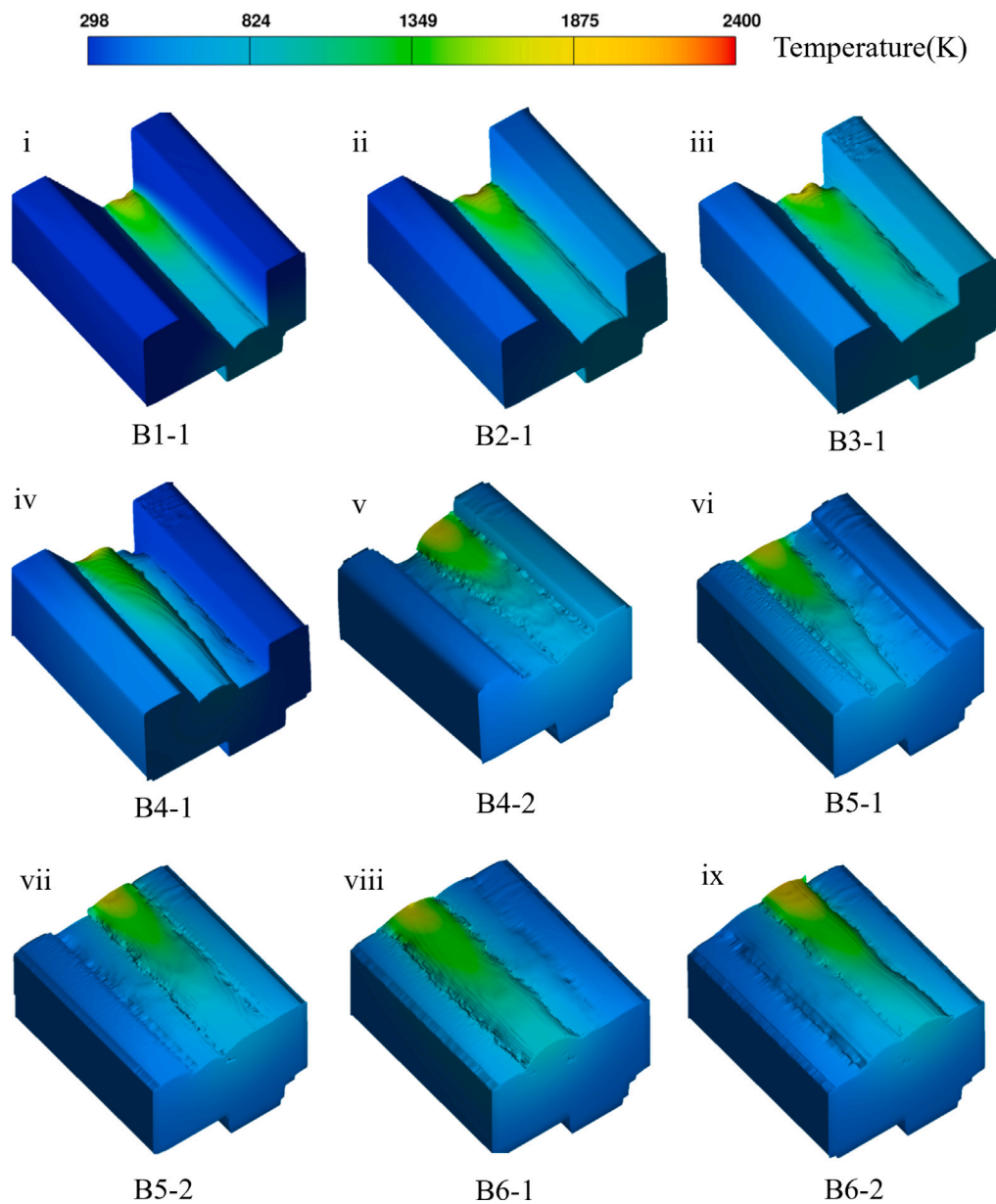


Fig. 10. Simulated results of the six-layer nine-pass welding task.

Table 4
Circle fitting results of the six-layer nine-pass welding simulation.

Weld bead	(x_c^p, y_c^p, r_c^p) (pixel)	Fitting error (pixel)	R^2
B1-1	(379.79, 474.27, 127.49)	0.85	0.99
B2-1	(348.33, 453.31, 189.57)	1.22	0.99
B3-1	(325.23, 450.42, 262.05)	1.96	0.97
B4-1	(193.76, 258.24, 147.49)	2.95	0.97
B4-2	(423.76, 409.33, 338.75)	1.34	0.99
B5-1	(155.48, 179.38, 147.78)	1.61	0.99
B5-2	(352.01, 177.65, 193.36)	1.82	0.99
B6-1	(138.48, 128.18, 193.24)	2.17	0.99
B6-2	(310.14, 99.21, 183.14)	1.85	0.99

the green circle is the fitted circle, and the green dot is the circle's center. The corresponding fitting errors calculated by Eq. (15) and the coefficient of determination calculated by Eq. (16) are summarized in Table 4.

The coefficients of determination indicate that the fitting results are satisfactory, and the fitting errors for the weld beads are minimal. The maximum fitting error is 2.95 pixels, approximately 0.15 mm. The results indicate that describing the weld bead contour with a circle is a reasonable and precise method. We also fit the welding bead profiles with a circle model in the simulated dataset. As shown in Fig. 12a, the fitting errors are distributed in a skewed normal distribution in the (0, 6) range, mainly concentrated around 1. The coefficients of determination shown in Fig. 12b are in the range of (0.8, 1), mainly concentrated around 0.98. These results indicate that the parameters (x_c^p, y_c^p, r_c^p) of the fitted circles effectively represent the geometric information of different weld beads in the simulated dataset.

3.3. Welding bead profile prediction

The model and training were implemented in PyTorch using a batch size of 32 and an Adam optimizer (Kingma and Ba, 2015) with a learning rate of 0.0005 on the loss function Eq. (26). The simulated dataset was split randomly into training and test sets in an 8:2 ratio.

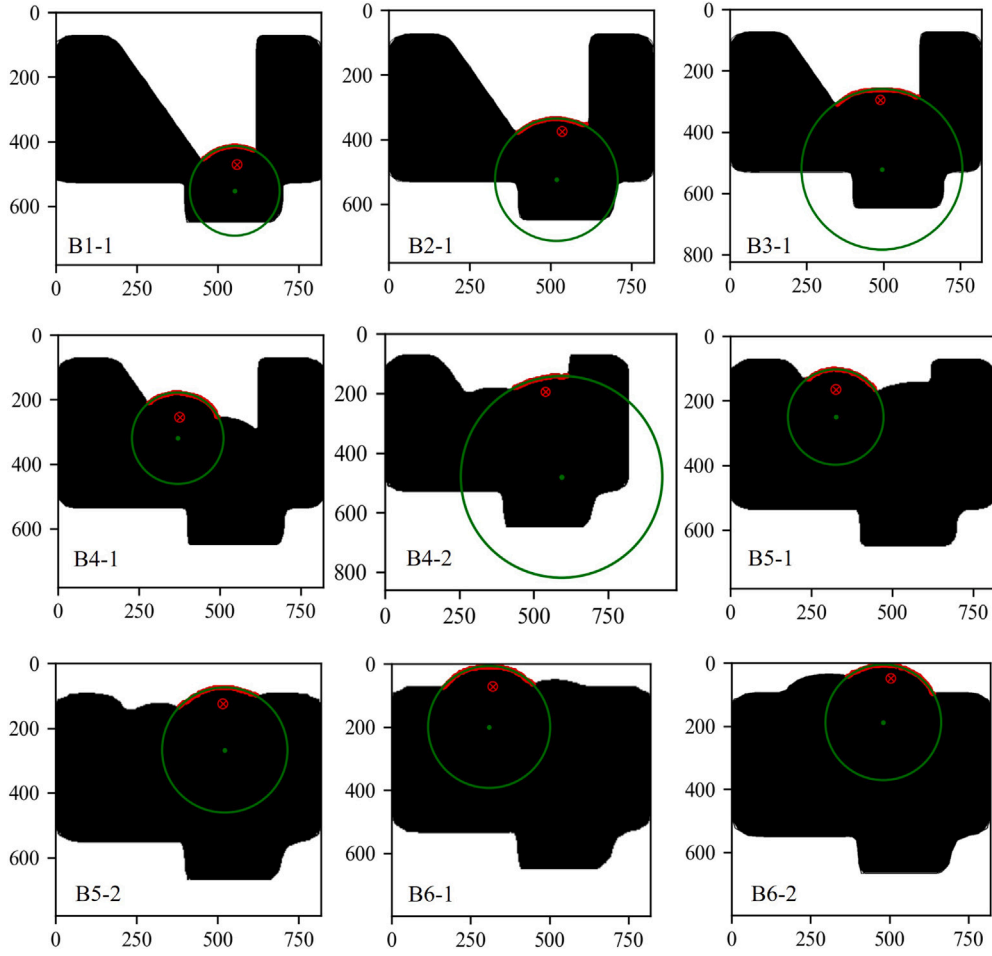


Fig. 11. Illustration of circle fitting on a six-layer nine-pass welding task. The green dot is the center of the circle, and the red $\otimes$ is the welding position.

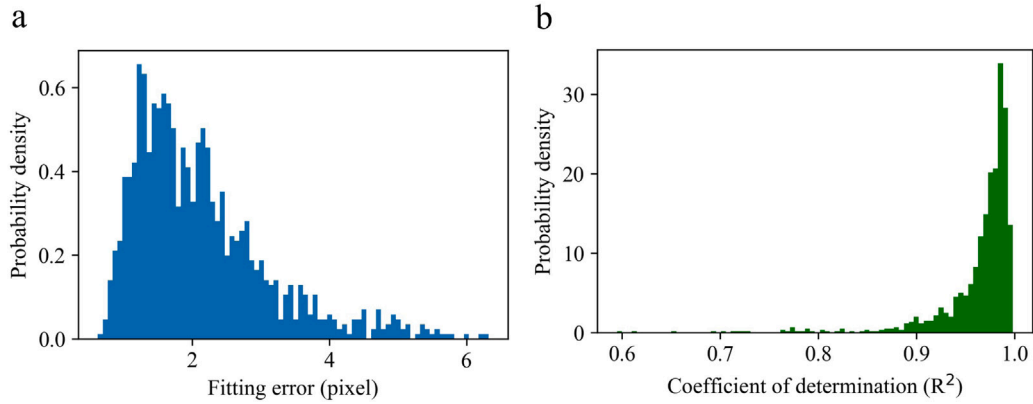


Fig. 12. Circle fitting results of weld bead profiles in the simulated dataset for deep learning. (a) Histogram of fitting errors. (b) Histogram of coefficients of determination.

We trained the model using a training dataset for 200 epochs and then verified its performance on the test dataset. The average prediction time is 15 ms on a NVIDIA GeForce RTX 3060 GPU. Figs. 13a&b show the position center errors and radius errors on the test dataset. The average position error is 22.62 pixels, and the average radius error is 21.15 pixels. The position and radius errors share a similar distribution, concentrating around 15 pixels (0.75 mm).

We further investigate the relationship between prediction errors and the radius size, and the normalized results are shown in Fig. 13c. The Pearson correlation coefficient is used to measure the strength of

correlation.

$$r_1 = \frac{\sum_{i=1}^{240} (e_{ci} - \bar{e}_c)(r_{ci}^p - \bar{r}_c^p)}{\sqrt{\sum_{i=1}^{240} (e_{ci} - \bar{e}_c)^2 \sum_{i=1}^{240} (r_{ci}^p - \bar{r}_c^p)^2}}, \quad (30)$$

$$r_2 = \frac{\sum_{i=1}^{240} (e_{ri} - \bar{e}_r)(r_{ci}^p - \bar{r}_c^p)}{\sqrt{\sum_{i=1}^{240} (e_{ri} - \bar{e}_r)^2 \sum_{i=1}^{240} (r_{ci}^p - \bar{r}_c^p)^2}}, \quad (31)$$

where e_{ci} , e_{ri} and r_{ci}^p represent the position error of the circle center, radius error, and radius of fitted circle on the test set as defined in

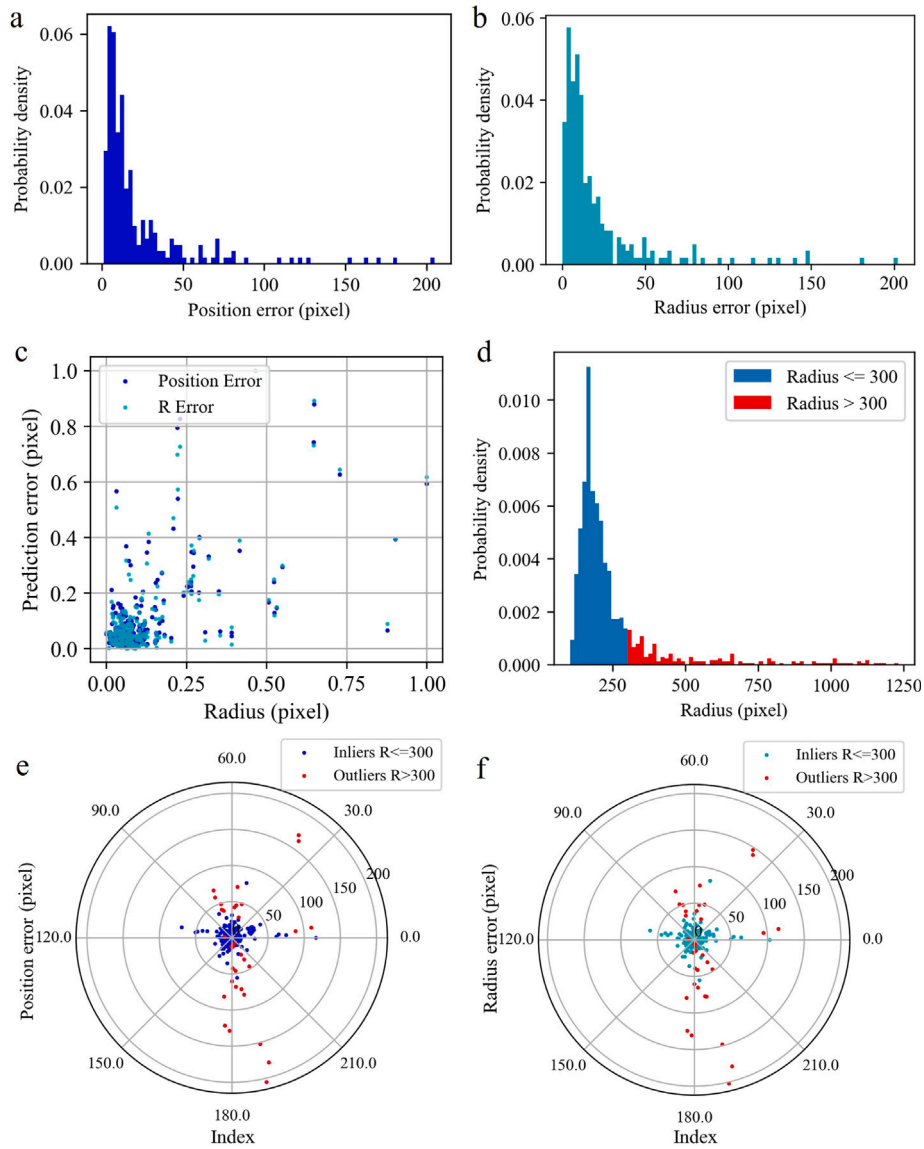


Fig. 13. Learning model evaluation on the test set. (a) Histogram of position errors. (b) Histogram of radius errors. (c) Relationship between prediction error and radius size. (d) Histogram of radius size distribution on the whole dataset. (e) Position errors in the polar coordinate system. (f) Radius errors in the polar coordinate system.

Eqs. (27), (28) and (23). $\bar{e}_{ci}$, $\bar{e}_{ri}$ and $\bar{r}_c^p$ are their respective means. r_1 and r_2 represent Pearson correlation coefficients of position and radius error against radius, respectively. r_1 is 0.59 and r_2 is 0.61, indicating a moderate positive correlation.

Fig. 13d shows the histogram of the radius size distribution in the whole dataset (1200 samples), where 193 data points' radius is more than 300 pixels, approximately 16.08% of the total dataset. Compared to the samples with a radius smaller than 300 ($r_c^p \leq 300$), these sparse samples with $r_c^p > 300$ have an extremely long tail, which may result in poor average prediction accuracy.

In the test dataset, we further investigate the prediction error distribution in the polar coordinate system as shown in Figs. 13e&f. The radial distance represents the magnitude of the errors, and the polar angle corresponds to the index of the sample in the test dataset. The test dataset's average position and radius errors are 22.62 pixels and 21.15 pixels, respectively, corresponding to 1.13 mm and 1.06 mm in actual size. There are 36 data points with $r_c^p > 300$, marked in red as the outliers. Excluding these outliers, the position and radius errors decrease to 14.61 pixels and 13.16 pixels, respectively, 0.73 mm and 0.66 mm in actual size. Additionally, the relative error of the radius

is 9.42%, which is lower than the total geometric error of 11.65% for the simplified characterization of weld bead geometry used in Wang et al. (2021). Employing a more appropriate method for weld bead characterization reduces the prediction error, improving accuracy. The results demonstrate that the proposed learning-based prediction model for the weld bead profile achieves high accuracy while maintaining low computational time.

3.4. Welding position prediction

To enhance the reusability and scalability of WeldPoseNet, we trained the encoder-decoder reconstruction and regression model in two stages using the dataset collected in the previous section. This dataset is identical to the one used for BeadProfileNet. However, while BeadProfileNet takes both the weld seam profile and the position of the molten droplet as input, WeldPoseNet only utilizes the weld seam profile. The model outputs the position of the molten droplet (which corresponds to the welding position) and the geometric parameters of the following weld bead. Furthermore, since the cross-section at each

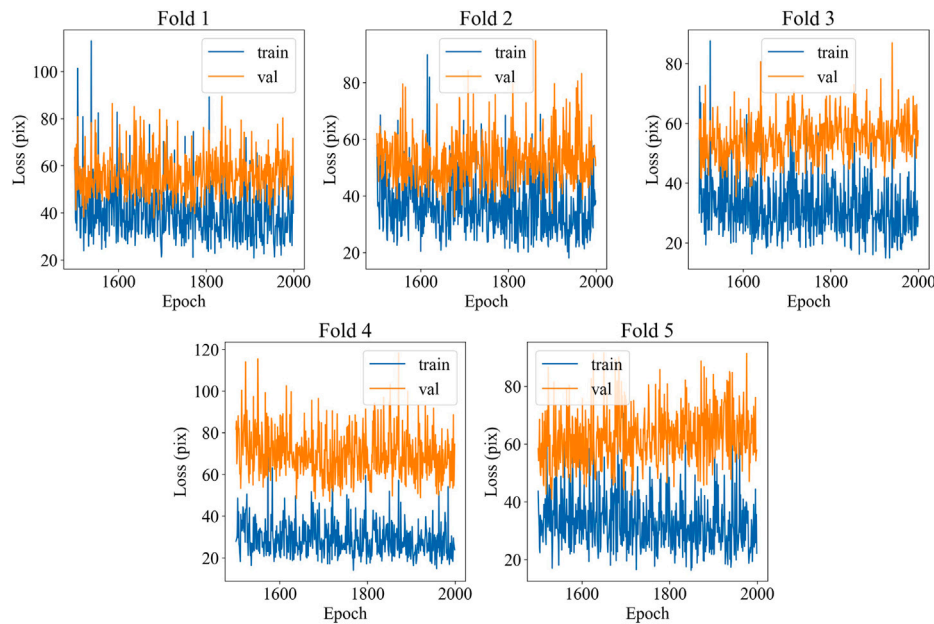


Fig. 14. Regression loss in five folds.

x -position of Bead 1 remains identical in the dataset, the total number of samples amounts to $1 + 200 \times 6 = 1201$.

In the first stage, the encoder–decoder reconstruction is a self-supervised learning. The dataset is randomly split into 8:2. It was trained with a batch size of 32 using an Adam optimizer with a learning rate of 0.001 on mean squared error loss (MSE Loss). The training epoch is set to 200, and we save the weights with the lowest training loss. We froze the encoder’s weights in the second stage and only trained the regression model. The batch size is 32, the optimizer is the Adam optimizer, and the learning rate is 0.0001. Due to the limited dataset size, a five-fold cross-validation approach is employed to maximize the use of available data for training the regression model. The training process is conducted for 2000 epochs.

The reconstruction loss of the encoder–decoder model on the validation dataset is 1.6896×10^{-4} , and the regression loss, as illustrated in Fig. 14 of the regression model across the five-fold cross-validation, is approximately 30 pixels, corresponding to 1.5 mm in actual size. These results demonstrate that the proposed WeldPoseNet achieves high accuracy in fitting simulated data.

3.5. Experimental validation of WeldPoseNet and BeadProfileNet

To validate the effectiveness of WeldPoseNet and BeadProfileNet, we collected real weld seam point cloud data and obtained a cross-section image of the weld seam through image transformation. The cross-section images were first fed into WeldPoseNet to predict the welding position. Subsequently, the cross-section image and the predicted welding position were input into BeadProfileNet to generate the predicted weld seam profile image. This predicted profile image was then fed back into WeldPoseNet, and the process was iteratively repeated until the weld seam was fully filled. The overall prediction workflow is illustrated in Fig. 15. It is worth noting that in practical applications, the ADAP framework offers considerable flexibility. The operator can choose to use WeldPoseNet to obtain a reference welding position based on real-time seam images, or alternatively, manually input different candidate positions into BeadProfileNet to compare the predicted bead geometries promptly. This functionality supports adaptive decision-making for optimal welding positioning.

Although some regions of the weld seam remain unfilled, this is primarily due to the dataset consisting mainly of data from intermediate filling layers, with a lack of data from the top layer. The

intermediate filling layers pose significant challenges in the MLMP welding process, and this study focuses primarily on them. Future work will focus on optimizing the planning of fully filled weld beads. Overall, the experimental results demonstrate that the proposed models can effectively simulate the MLMP welding process, enabling real-time welding position planning for various weld seams during thick plate welding. Additionally, input images may contain noise or outliers due to spatter or other welding defects, such as the noise points shown in Fig. 15. The existing method, which calculates welding positions based on the curve slope, may fail in such cases (He et al., 2020). However, the deep learning-based models presented in this study demonstrate excellent robustness in handling these challenges.

4. Conclusion, limitations & future work

This study introduces the Adaptive and Dynamic Arc Padding (ADAP) framework, which seamlessly integrates three key components: BeadProfileNet, WeldPoseNet, and an Arc Model for accurate representation of weld bead profiles. BeadProfileNet is a robust model for predicting weld bead geometry, while WeldPoseNet is designed to estimate the optimal welding torch position. Together, these components enable real-time prediction and adaptive control of weld bead formation in the Multi-Layer-Multi-Pass (MLMP) welding, significantly enhancing process efficiency, precision, and automation.

By combining high-fidelity Flow-3D simulations with metallographic validation, we construct a physically consistent and data-rich MLMP welding dataset. This not only reduces the cost and complexity of traditional experimental trials but also allows detailed visualization of key physical fields such as temperature, fluid flow, and stress, which facilitates comprehensive analysis and validation of welding behavior. Meanwhile, the simulated cross-sectional bead profiles serve as a reliable and scalable data source for training deep learning models.

Compared to conventional shape approximations (e.g., rectangular, trapezoidal, or parabolic models), our method offers a more accurate and compact representation of complex weld seam geometries while maintaining high computational efficiency. This modeling technique serves as the geometric backbone of the ADAP framework, facilitating robust data extraction for learning-based approaches.

In addition, this study proposed a model-driven predictive framework, consisting of BeadProfileNet and WeldPoseNet. BeadProfileNet

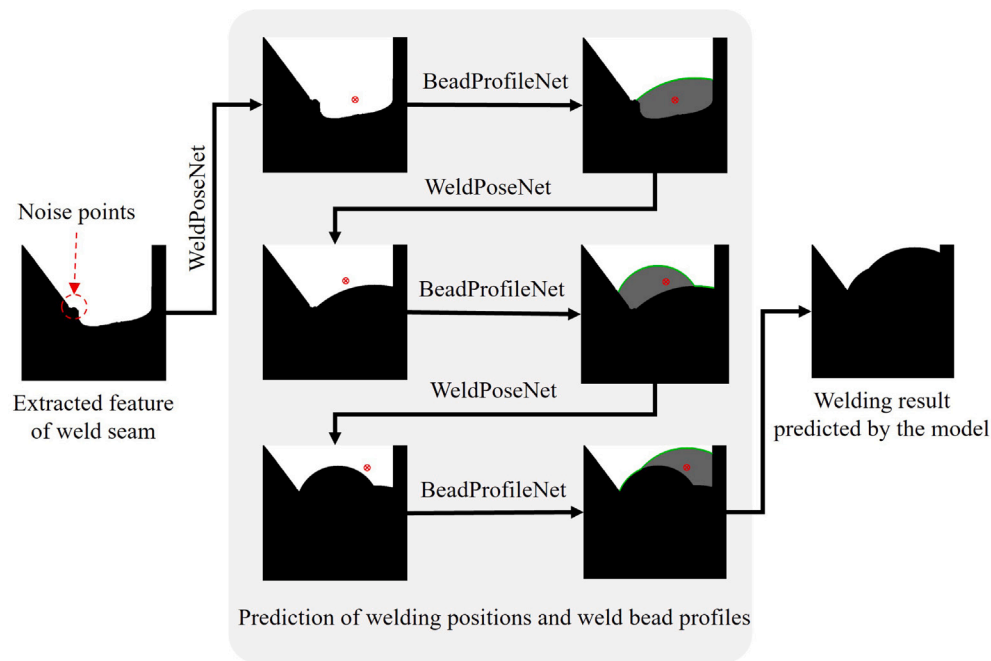


Fig. 15. Experimental validation flowchart of WeldPoseNet and BeadProfileNet.

predicts weld bead geometry with an average error of 0.73 mm in position and 0.66 mm in radius, and operates within 15 ms on GPU hardware—far exceeding the speed of conventional CFD methods. WeldPoseNet complements this by predicting torch positions within 0.025 s, enabling real-time trajectory adjustments and autonomous filler wire placement. This transition from rule-based heuristics to data-driven control significantly advances the field of intelligent robotic welding.

Beyond its immediate application in robotic welding, the ADAP framework advances data-driven modeling in manufacturing by accelerating MLMP welding process design cycles and facilitating rapid exploration of welding parameter spaces. This capability enables researchers and practitioners to optimize welding conditions, investigate critical factors that influence weld seam formation, and improve overall weld quality. These innovations collectively lay the groundwork for developing intelligent, autonomous welding systems.

Despite its promising performance, the current study has several limitations. The inherent complexity of MLMP welding means that our fluid simulations cannot fully capture all aspects of the physical processes involved, leading to discrepancies between simulated weld bead geometries and actual experimental outcomes. Variability in welding parameters, such as voltage, current, and the thermal properties of the workpiece and filler material, further complicates the accuracy of simulations. Additionally, to generate a large volume of simulation data, the employed setup utilized a diagonally moving heat source, which resulted in varying droplet positions across different weld seam cross-sections. While this approach approximates diverse welding conditions, it may not capture all nuances of weld bead formation. Moreover, the limited size of our current dataset has led to reduced prediction accuracy for weld beads with larger radii and suboptimal welding position predictions. Consequently, the ADAP framework is still nascent, and its performance in practical welding experiments requires further refinement.

Future research will focus on several critical areas to enhance and validate the ADAP framework, including (a) Enhanced Simulation Fidelity, (b) Dataset Expansion and Generalization, (c) Empirical Validation on Robotic Systems, and (d) Integration into Industrial Applications. We will refine our welding simulation to enhance model accuracy regarding the effects of key parameters, including torch orientation, welding speed, current, and voltage on weld bead geometry.

Additionally, we plan to investigate the potential impact of thermal conductivity, liquidus temperature, and other material properties on simulation outcomes. Improved simulation fidelity will help bridge the gap between simulated and real-world welding outcomes. Future efforts will be made to expand the simulation dataset by incorporating a wider range of welding parameters and groove geometries. A more extensive and diverse dataset will improve the robustness and generalization capabilities of both BeadProfileNet and WeldPoseNet. A primary objective for future work is to empirically validate the ADAP framework on an integrated robotic welding system. In a complete system, a 3D scanner mounted above the workpiece will capture the real-time 3D geometry of the weld seam. Cross-sectional data extracted from these scans will be processed using the Arc Model to derive weld bead geometric parameters, which will then be input to the prediction models. An evaluation module will assess the predicted weld quality, enabling the welding robot to adjust its motion trajectory dynamically for optimal filler wire placement. Ultimately, the goal is to integrate the ADAP framework into a fully automated robotic welding system for thick plates. This system will leverage real-time feedback from 3D scanning and machine learning-driven predictions to achieve high-quality, efficient welding, addressing both the practical demands of industrial manufacturing and the evolving challenges of MLMP welding.

CRediT authorship contribution statement

He Wang: Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis, Data curation. **Sen Li:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Xiaobo Liu:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Chengxiao Dong:** Visualization, Validation, Investigation, Formal analysis. **Fang Wan:** Writing – review & editing, Visualization, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Science, Technology, and Innovation Commission of Shenzhen Municipality under Grant JSGG20220831110002004, the National Natural Science Foundation of China under Grant 62206119, and Shenzhen Long-Term Support for Higher Education at SUSTech under Grant 20231115141649002.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.aiopen.2025.10.003>.

Data availability

All data and codes related to this work are hosted on the following project website: <https://github.com/ancorasir/WeldingADAP>.

References

- Akbari, P., Ogoke, F., Kao, N.Y., Meidani, K., Yeh, C.Y., Lee, W., Farimani, A.B., 2022. MeltPoolNet: Melt pool characteristic prediction in metal additive manufacturing using machine learning. *Addit. Manuf.* 55, 102817. <http://dx.doi.org/10.1016/j.addma.2022.102817>.
- Cao, Z., Yang, Z., Chen, X., 2004. Three-dimensional simulation of transient GMA weld pool with free surface. *Weld. J.* 83, 169–S.
- Cho, M., Farson, D., 2007. Simulation study of a hybrid process for the prevention of weld bead hump formation. *Weld. J.* 86 (9), 253–S.
- Cho, J., Na, S.J., 2009. Three-dimensional analysis of molten pool in GMA-laser hybrid welding. *Weld. J.* 88 (2), 35–43.
- Cho, D.W., Na, S.J., 2015. Molten pool behaviors for second pass V-groove GMAW. *Int. J. Heat Mass Transfer* 88, 945–956. <http://dx.doi.org/10.1016/j.ijheatmasstransfer.2015.05.021>.
- Cho, D., Na, S.J., Cho, M., Lee, J., 2013. A study on V-groove GMAW for various welding positions. *J. Mater. Process. Technol.* 213 (9), 1640–1652. <http://dx.doi.org/10.1016/j.jmatprotec.2013.02.015>.
- Dai, D., Gu, D., 2014. Thermal behavior and densification mechanism during selective laser melting of copper matrix composites: Simulation and experiments. *Mater. Des.* 55, 482–491. <http://dx.doi.org/10.1016/j.matdes.2013.10.006>.
- Deng, J., Xu, Y., Zuo, Z., Hou, Z., Chen, S., 2019. Bead geometry prediction for multi-layer and multi-bead wire and arc additive manufacturing based on XGBoost. In: Chen, S., Zhang, Y., Feng, Z. (Eds.), *Transactions on Intelligent Welding Manufacturing*. Springer, Singapore, pp. 125–135. http://dx.doi.org/10.1007/978-981-13-8668-8_7.
- Flow Science Japan, 2023. *FSJ module manual*. FLOW-3D Software Documentation.
- He, Y., Cai, R., Dai, F., Yu, Z., Deng, Y., Deng, J., Wang, Z., Ma, G., Zhong, W., 2024a. A unified framework based on semantic segmentation for extraction of weld seam profiles with typical joints. *J. Manuf. Process.* 131, 2275–2287. <http://dx.doi.org/10.1016/j.jmapro.2024.10.042>.
- He, Y., Li, D., Pan, Z., Ma, G., Yu, L., Yuan, H., Le, J., 2020. Dynamic modeling of weld bead geometry features in thick plate GMAW based on machine vision and learning. *Sensors* 20 (24), 7104. <http://dx.doi.org/10.3390/s20247104>.
- He, Y., Ma, G., Chen, S., 2021. Autonomous decision-making of welding position during multipass GMAW with T-joints: A Bayesian network approach. *IEEE Trans. Ind. Electron.* 69 (4), 3909–3917. <http://dx.doi.org/10.1109/TIE.2021.3076710>.
- He, Y., Yu, Z., Deng, Y., Deng, J., Cai, R., Wang, Z., Tu, W., Zhong, W., 2024b. AHP-based welding position decision and optimization for angular distortion and weld collapse control in T-joint multipass GMAW. *J. Manuf. Process.* 121, 246–259. <http://dx.doi.org/10.1016/j.jmapro.2024.05.014>.
- Hemmasian, A., Ogoke, F., Akbari, P., Malen, J., Beuth, J., Farimani, A.B., 2023. Surrogate modeling of melt pool temperature field using deep learning. *Addit. Manuf. Lett.* 5, 100123. <http://dx.doi.org/10.1016/j.addlet.2023.100123>.
- Hirt, C.W., Nichols, B.D., 1981. Volume of fluid (VOF) method for the dynamics of free boundaries. *J. Comput. Phys.* 39 (1), 201–225. [http://dx.doi.org/10.1016/0021-9991\(81\)90145-5](http://dx.doi.org/10.1016/0021-9991(81)90145-5).
- Kingma, D.P., Ba, J.L., 2015. Adam: A method for stochastic optimization. In: *Proceedings of the 3rd International Conference on Learning Representations*. ICLR, <https://arxiv.org/abs/1412.6980>.
- Lee, S.H., Kim, E., Park, J., Choi, J., 2018. Numerical analysis of thermal deformation and residual stress in automotive muffler by MIG welding. *J. Comput. Des. Eng.* 5 (4), 382–390. <http://dx.doi.org/10.1016/j.jcde.2018.05.001>.
- Li, Y., Sun, Y., Han, Q., Zhang, G., Horváth, I., 2018. Enhanced beads overlapping model for wire and arc additive manufacturing of multi-layer multi-bead metallic parts. *J. Mater. Process. Technol.* 252, 838–848. <http://dx.doi.org/10.1016/j.jmatprotec.2017.10.017>.
- Lin, Q., Li, X., Simpson, S., 2001. Metal transfer measurements in gas metal arc welding. *J. Phys. D: Appl. Phys.* 34 (3), 347. <http://dx.doi.org/10.1088/0022-3727/34/3/317>.
- Liu, X., Han, X., Guo, N., Wan, F., Song, C., 2023. Bio-inspired proprioceptive pouch of a soft finger with inner-finger kinesthetic perception. *Biomimetics* 8 (6), 501. <http://dx.doi.org/10.3390/biomimetics8060501>.
- Lu, M., Kou, S., 1988. Power and current distributions in gas tungsten arcs. *Weld. J.* 67, 2.
- Lu, X., Zhang, G., Li, J., Cervera, M., Chiumenti, M., Chen, J., Lin, X., Huang, W., 2021. Simulation-assisted investigation on the formation of layer bands and the microstructural evolution in directed energy deposition of Ti6Al4V blocks. *Virtual Phys. Prototyp.* 16 (4), 387–403. <http://dx.doi.org/10.1080/17452759.2021.1942077>.
- Ma, D., Jiang, P., Shu, L., Gong, Z., Wang, Y., Geng, S., 2024. Online porosity prediction in laser welding of aluminum alloys based on a multi-fidelity deep learning framework. *J. Intell. Manuf.* 35 (1), 55–73. <http://dx.doi.org/10.1007/s10845-022-02033-9>.
- Madsen, O., Bro Sørensen, C., Larsen, R., Overgaard, L., Jacobsen, N.J., 2002. A system for complex robotic welding. *Ind. Robot.: An Int. J.* 29 (2), 127–131. <http://dx.doi.org/10.1108/01439910210419132>.
- Manokruang, S., Vignat, F., Museau, M., Limousin, M., 2021. Process parameters effect on weld beads geometry deposited by wire and arc additive manufacturing (WAAM). *Adv. Mech. Des. Eng. Manuf.* 3, 9–14. http://dx.doi.org/10.1007/978-3-030-70566-4_3.
- Matsushita, Y., Tran, D.T., Yamazoe, H., Lee, J.H., 2021. Recent use of deep learning techniques in clinical applications based on gait: a survey. *J. Comput. Des. Eng.* 8 (6), 1499–1532. <http://dx.doi.org/10.1093/jcde/qwab054>.
- Raouf, I., Lee, H., Noh, Y.R., Youn, B.D., Kim, H.S., 2022. Prognostic health management of the robotic strain wave gear reducer based on variable speed of operation: A data-driven via deep learning approach. *J. Comput. Des. Eng.* 9 (5), 1775–1788. <http://dx.doi.org/10.1093/jcde/qwac091>.
- Shin, H.S., Boo, S.H., 2022. Welding simulation using a reduced order model for efficient residual stress evaluation. *J. Comput. Des. Eng.* 9 (4), 1196–1213. <http://dx.doi.org/10.1093/jcde/qwac047>.
- Suryakumar, S., Karunakaran, K., Bernard, A., Chandrasekhar, U., Raghavender, N., Sharma, D., 2011. Weld bead modeling and process optimization in hybrid layered manufacturing. *Comput.-Aided Des.* 43 (4), 331–344. <http://dx.doi.org/10.1016/j.cad.2011.01.006>.
- Wang, S., Zhu, L., Fuh, J.Y.H., Zhang, H., Yan, W., 2020. Multi-physics modeling and Gaussian process regression analysis of cladding track geometry for direct energy deposition. *Opt. Lasers Eng.* 127, 105950. <http://dx.doi.org/10.1016/j.optlaseng.2019.105950>.
- Wang, Z., Zimmer-Chevret, S., Léonard, F., Abba, G., 2021. Prediction of bead geometry with consideration of interlayer temperature effect for CMT-based wire-arc additive manufacturing. *Weld. World* 65 (12), 2255–2266. <http://dx.doi.org/10.1007/s40194-021-01192-2>.
- Xiong, J., Zhang, G., Gao, H., Wu, L., 2013. Modeling of bead section profile and overlapping beads with experimental validation for robotic GMAW-based rapid manufacturing. *Robot. Comput.-Integr. Manuf.* 29 (2), 417–423. <http://dx.doi.org/10.1016/j.rcim.2012.09.011>.
- Xu, G., Cao, Q., Hu, Q., Zhang, W., Liu, P., Du, B., 2016. Modelling of bead hump formation in high speed gas metal arc welding. *Sci. Technol. Weld. Join.* 21 (8), 700–710. <http://dx.doi.org/10.1080/13621718.2016.1146427>.
- Xu, G., Wang, J., Li, P., Zhu, J., Cao, Q., 2018. Numerical analysis of heat transfer and fluid flow in swing arc narrow gap GMA welding. *J. Mater. Process. Technol.* 252, 260–269. <http://dx.doi.org/10.1016/j.jmatprotec.2017.09.030>.
- Yang, C., Ye, Z., Chen, Y., Zhong, J., Chen, S., 2014. Multi-pass path planning for thick plate by DSAW based on vision sensor. *Sensor Rev.* 34 (4), 416–423. <http://dx.doi.org/10.1108/SR-04-2013-649>.
- Zhan, X., Zhang, D., Liu, X., Chen, J., Wei, Y., Zhou, J., Liu, R., 2017. Comparison between weave bead welding and multi-layer multi-pass welding for thick plate invar steel. *Int. J. Adv. Manuf. Technol.* 88, 2211–2225. <http://dx.doi.org/10.1007/s00170-016-8926-4>.
- Zhang, H., Lu, H., Cai, C., Chen, S., 2011. Robot path planning in multi-pass weaving welding for thick plates. In: Tarn, T.J., Chen, S.B., Fang, G. (Eds.), *Robotic Welding, Intelligence and Automation*. In: *Lecture Notes in Electrical Engineering*, vol. 88, Springer, Berlin, Heidelberg, pp. 351–359. http://dx.doi.org/10.1007/978-3-642-19959-2_43.